

SOFTWARE TOOL ARTICLE

CADEMAS-ML: A Software Tool for Context-Aware Cooperative and Explainable Automated Decision-Making

Pavel Novoa-Hernández^a and David A. Pelta^b

^aDept. Computer Engineering and Systems, Universidad de La Laguna, San Cristóbal de La Laguna, Spain; ^bDept. Computer Science and Artificial Intelligence, Universidad de Granada, Granada, Spain

ARTICLE HISTORY

Compiled October 2, 2026

ABSTRACT

~~Complex real-world decision problems often involve multiple decision-making actors that must combine their assessments while accounting for the organizational, operational, and strategic context surrounding the problem.~~ Complex organizational decisions often combine assessments from multiple actors with the context surrounding the problem. The Cooperative Automated Decision-Making Systems (CADEMAS) framework was recently proposed to represent this form of cooperative and context-aware decision-making. However, previous CADEMAs instantiations cover only specific configurations of a framework that allows substantially broader combinations of automated decision components, contextual models, and aggregation mechanisms. ~~This flexibility, while valuable for practical applications, also makes the framework difficult to operationalize without dedicated software support.~~ This breadth makes the framework difficult to operationalize without dedicated software support. This paper presents CADEMAs-ML, a software tool that implements a configurable class of CADEMAs systems in which automated decision-making components are machine learning models and contextual knowledge is represented through fuzzy logic. ~~The tool supports the integration and weighting of multiple predictive models, configurable fuzzy context models, alternative aggregation operators, and an adjustable trade-off between predictive risk and contextual suitability. CADEMAs-ML also incorporates explainability and robustness analysis to support the inspection and assessment of resulting decisions.~~ It integrates weighted predictive models with configurable fuzzy contexts and alternative aggregation operators. Explainability and robustness analyses support inspection of the resulting decisions. An employee attrition case study illustrates how the same predictive evidence can lead to different prioritizations under alternative contextual scenarios.

KEYWORDS

automated decision-making systems; fuzzy decision making; context-aware decision support; machine learning; prioritization; cooperative systems

1. Introduction

Decision-making in real-world organizations is rarely a matter of applying a single analytical model to a well-defined set of inputs. Complex decisions often involve multiple stakeholders, organizational units, or analytical components that contribute different pieces of evidence to the same problem. ~~These contributions may reflect different objectives, information sources, areas of expertise, or levels of reliability, and therefore need to be coordinated before they can support a coherent course of action.~~ Their objectives, information, expertise, and reliability may differ. Coordination is therefore needed before the evidence can support a coherent course of action.

This perspective becomes particularly important when automated decision-making is used to support prediction, classification, or risk assessment. ~~An automated decision-making (ADM) component can produce a probability, rating, recommendation, or risk score, but such an output does not by itself determine how the result should be acted upon. The appropriate course of action may depend on factors that are external to the predictive model, including organizational priorities, available resources, departmental objectives, regulatory constraints, or the strategic conditions under which the decision is made. In other words, decisions are taken in a “context” and not in vacuum.~~ An automated decision-making (ADM) component may produce a probability, rating, recommendation, or risk score. Acting on that output also requires attention to organizational priorities, available resources, regulations, and strategy. Decisions are thus made within a context.

~~Besides this, the simultaneous use of different ADM components may provide complementary views of the cases under consideration, while the final decision may also need to reflect contextual knowledge provided by domain experts or organizational policies.~~ Several ADM components can provide complementary views of the same cases. Domain knowledge and organizational policies then guide how those views inform the final decision.

These considerations motivated the Cooperative Automated Decision-Making Systems (CADEMAS) framework proposed by Novoa-Hernández, Pelta, Godz, Verdery, and Buendia-Carrillo (2026). ~~CADEMAS provides a conceptual framework for coordinating heterogeneous automated decision-making components while explicitly incorporating the context in which their outputs are interpreted. Its central principle is to distinguish between the generation and integration of decision evidence and the contextual evaluation of that evidence. Multiple automated decision units can therefore contribute to a collective assessment, while a separate contextual layer represents organizational priorities, policies, or scenario-dependent considerations. The resulting decision is obtained by combining these two sources of information rather than treating the predictive assessment as an isolated or definitive outcome.~~ CADEMAs coordinates heterogeneous ADM components and incorporates the context in which their outputs

are interpreted. It connects an integrated assessment with a separate representation of organizational priorities, policies, or scenarios. Section 2.1 formalizes this architecture.

~~The work is inspired in the development of cooperative and multi-agent decision-support systems, in which heterogeneous decision components can contribute to a common decision process~~ The framework builds on cooperative and multi-agent decision-support systems, where heterogeneous components contribute to a common process (Bui & Lee, 1999; Vahidov & Fazlollahi, 2004) ~~and also, in multiperson and group decision-support research, where heterogeneous sources of information and preferences have traditionally been combined through aggregation, negotiation, and consensus-seeking mechanisms.~~ It also draws on multiperson and group decision support, which combines heterogeneous information and preferences through aggregation, negotiation, and consensus mechanisms (Herrera-Viedma, Herrera, & Chiclana, 2002; Salo, 1995). ~~Similar principles can be extended to automated settings, where the cooperating components could be software-based decision units rather than human decision makers.~~ In an automated setting, software-based decision units assume the cooperating roles.

~~A particular scenario where a CADEMAs-like tool is relevant is the following: suppose that you have a set of entities and some limited intervention budget. The question is how to prioritize the entities for receiving an intervention?~~ One relevant scenario involves a set of entities and a limited intervention budget. The decision problem is to establish which entities should receive attention first.

~~Examples of this scenario are~~ Examples include attrition risk prevention Godz, Novoa-Hernández, and Pelta (2026), ~~small and medium size companies financial rescue~~ financial rescue for small and medium-sized companies Navarro-Galera, Lara-Rubio, Novoa-Hernández, and Cruz Corona (2025), ~~students dropout prevention~~ student-dropout prevention (BUSCAR REF), ~~and so on.~~

~~Using CADEMAs in these problems means that first, a risk is assessed for every entity using different ADMs (implementing machine learning techniques) and second, the suitability of the entity to a given context is also calculated. For example, in a context of economic crisis you may prioritize a company with a high number of employees (or with the lower); in a context of growing, you may prioritize companies with a high expected impact. Such contexts can be modeled, for example, with fuzzy rules as proposed in~~ In these problems, CADEMAs combines the assessments of multiple ADMs with the suitability of each entity for the decision context. Section 3 develops this connection for prioritization. Fuzzy rules can represent such contexts (Lamata, Pelta, & Verdegay, 2020; Pérez-Cañedo, Porras, Pelta, & Verdegay, 2023), ~~allowing to combine quantitative information with expert knowledge and imprecise or linguistic assessments,~~ combining quantitative information with expert knowledge and linguistic assessments.

~~But also, users may need to understand why a particular entity receives a given~~

prediction and how changes in the underlying information affect the result. Explainable artificial intelligence has consequently become an important area of research for the practical use of machine-learning models, particularly when predictions contribute to decisions with organizational or human consequences. Users also need to understand individual predictions, especially when they influence organizational or human outcomes (Barredo Arrieta et al., 2020). Model-agnostic approaches such as SHAP provide local feature attributions that can help users inspect the factors contributing to individual predictions. Local feature attributions can reveal the factors associated with each prediction (Lundberg et al., 2020; Lundberg & Lee, 2017).

Robustness analysis is another required capability for context-aware decision support. In CADEMAs, the final prioritization depends not only on predictive evidence but also on contextual definitions, aggregation mechanisms, and the relative importance assigned to predictive and contextual information. Changes in these elements may alter the ordering of cases even when the underlying predictive models remain unchanged. Allowing users to examine such changes can provide information about the stability of the resulting decisions and expose cases whose prioritization is particularly sensitive to the adopted policy configuration. The ordering of cases may also change with the contextual definition, aggregation mechanism, or balance assigned to predictive and contextual information. Robustness analysis reveals stable priorities and cases that are sensitive to these policy choices (Barredo Arrieta et al., 2020).

To the best of our knowledge, there is no integrated tool that allows to deal with prioritization problems in the way described before. To the best of our knowledge, no integrated tool supports this complete prioritization workflow.

In this scenario, the main aim is to present CADEMAs-ML, a software tool designed to engineer a configurable class of CADEMAs systems (to deal with prioritization problems) in which automated decision-making components are implemented as machine-learning models and contextual knowledge is represented through fuzzy rules. It provides configurable mechanisms for representing and aggregating contextual conditions, and combines predictive risk with contextual suitability through an explicit trade-off mechanism. In addition to these core CADEMAs capabilities, this work proposes explainability and robustness analysis as natural extensions of the framework. Both capabilities are incorporated into CADEMAs-ML. This paper presents CADEMAs-ML to fill that gap. The tool implements a configurable class of CADEMAs systems for prioritization, with machine-learning ADM units and fuzzy contextual knowledge. It also extends the workflow with case-level explanation and robustness analysis.

The remainder of the paper is organized as follows. Section 2 reviews the CADEMAs framework and related work on cooperative and fuzzy decision-support systems. Section 3 presents the architecture and main components of CADEMAs-ML. Section 4 reports the outputs produced by its modules. Both sections use an employee attrition

~~case study to illustrate the functionality and operation of the tool.~~ Section 3 formalizes its specialization for prioritization. Section 4 presents the software architecture and its main functionalities through an employee-attribution case study. Finally, Section 5 summarizes the main findings and discusses the implications of the proposed tool and possible directions for future work.

2. Background and related works

This section sets out the conceptual foundations on which CADEMAs-ML builds. We first revisit the CADEMAs framework in its general form (Section 2.1). We then situate the tool among related cooperative-decision frameworks and application-oriented decision-support systems, clarifying the gap it is designed to fill (Section 2.2).

2.1. CADEMAs framework

As introduced in Section 1, CADEMAs (Novoa-Hernández et al., 2026) provides a general conceptual and architectural framework for structuring cooperation among heterogeneous Automated Decision-Making (ADM) units. Its scope is not restricted to prediction. ~~Organizational decisions may involve several decision-producing components, including classifiers, optimization models, simulation components, rule-based systems, large language model agents, or institutional procedures. These components may differ in their assumptions, data access, output types, and reliability. Rather than prescribing a particular learning method, rule formalism, or coordination protocol, CADEMAs defines how heterogeneous decision-producing components can contribute to a common decision process and how their outputs can subsequently be considered in the context of the problem.~~ The participating components may be classifiers, optimization or simulation models, rule-based systems, software agents, or institutional procedures. Their assumptions, data, outputs, and reliability may differ. CADEMAs defines how they contribute to a common process and how context informs the resulting decision without imposing a particular learning method or coordination protocol.

~~A central premise of CADEMAs is that an ADM unit does not necessarily produce a complete operational decision. Instead, it may provide a *partial decision signal*, such as a score, prediction, rule activation, explanation, or suggested action. The semantics of these signals may also differ across units. They therefore need to be integrated before a final decision is obtained and may subsequently be adjusted according to the relevant context.~~ An ADM unit may contribute a *partial decision signal*, such as a score, prediction, rule activation, explanation, or suggested action. Signals with different semantics are integrated and then evaluated in their context. Formally, a

CADEMAS instance is represented by the tuple

$$\mathcal{S} = \langle \mathcal{IM}, \mathcal{A}, \mathcal{OM}, \mathcal{C}, \mathcal{T} \rangle, \quad (1)$$

where $\mathcal{IM}$ is the Input Manager, $\mathcal{A}$ is the set of ADM units, $\mathcal{OM}$ is the Decision Integration Layer, $\mathcal{C}$ is the Contextual Modulator, and $\mathcal{T}$ is the universe of admissible data types and representations.

The Input Manager $\mathcal{IM}$ routes, transforms, and validates the information entering the architecture. It distinguishes training or calibration data, which are used offline to construct or update the internal models of the ADM units, from decision-case data provided when a new case is evaluated. The set of ADM units is denoted by

$$\mathcal{A} = \{ADM_1, \dots, ADM_n\}. \quad (2)$$

Each ADM_i receives a data representation $d_i \in \mathcal{T}$ and produces an output $o_i \in \mathcal{T}$. Cooperation among these units may be *indirect*, with each unit contributing an independent signal, or *direct*, with units exchanging intermediate representations. CADEMAs allows for both forms without imposing a specific communication protocol.

Given the heterogeneous collection $\mathcal{O} = \{o_1, \dots, o_n\}$, the Decision Integration Layer $\mathcal{OM}$ synthesizes a unified decision object D through a mapping

$$\Phi : \mathcal{O} \rightarrow D.$$

This layer addresses the harmonization of ADM outputs and their subsequent aggregation or synthesis. ~~The resulting D is a *raw* decision proposal: the outputs have been brought into a common decision representation, but the proposal has not yet been adjusted to the context in which it will be used. The Contextual Modulator provides this additional step by modifying the integrated proposal according to the relevant contextual conditions.~~ The resulting D is a *raw* proposal in a common representation. The Contextual Modulator then adjusts it to the relevant conditions. Formally,

$$M : D \times \mathcal{C} \rightarrow D, \quad D^{\text{final}} = M(D^{\text{raw}}, \mathcal{C}), \quad (3)$$

where $\mathcal{C}$ may encode regulatory requirements, organizational policies, resource constraints, or other scenario-dependent conditions. ~~The framework does not prescribe a single representation for contextual information. Depending on the application, it may be expressed through crisp rules, utility functions, regulatory constraints, fuzzy propositions, or other complementary mechanisms.~~ Its representation may use rules, utility functions, constraints, fuzzy propositions, or other mechanisms suited to the application.

The predictive and fuzzy specialization implemented by CADEMAs-ML, including cooperative risk scores, fuzzy context alignment, and an explicit trade-off between these components, is described in ~~Section 3~~Section 3. ~~The general CADEMAs architecture remains open with respect to the type of ADM units, the semantics of their outputs, the harmonization and aggregation mechanisms implemented by Φ , and the way contextual information is incorporated through M .~~ The general architecture leaves the ADM types, output semantics, integration mapping Φ , and contextual modulation M open to specialization.

This flexibility is reflected in the applications reported to date. The original formulation illustrated the framework with an SME insolvency-prediction scenario in which heterogeneous decision units were combined with contextual constraints (Novoa-Hernández et al., 2026). Godz et al. (2026) subsequently instantiated CADEMAs for employee-attrition prevention, treating departmental machine-learning models as co-operating ADM units and encoding organizational scenarios through a fuzzy context layer. Building on this machine-learning pathway, Novoa-Hernández, Godz, and Pelta (2026) examined the effect of different aggregation operators on prioritization, comparing linear, weighted-geometric, and minimum operators and analyzing how their aggregation behavior affects the interaction between predictive evidence and contextual requirements. More recently, Vazquez-Sánchez, Buendía-Carrillo, and Corona (2026) proposed a cooperative ADM strategy with fuzzy contextual modulation for SME insolvency prediction, combining heterogeneous survival models, a contextual modulator, and explainable-AI support.

~~These studies illustrate different ways in which the CADEMAs architecture can be instantiated, but each requires a specific implementation for the domain and configuration considered. A reusable software environment is therefore useful for constructing, inspecting, comparing, and reproducing CADEMAs configurations. CADEMAs-ML addresses this need by providing such an environment for a machine-learning-based specialization of the framework.~~ Together, these studies show the range of possible CADEMAs configurations. CADEMAs-ML turns their shared machine-learning pathway into a reusable environment for constructing, inspecting, comparing, and reproducing such configurations.

2.2. *Related frameworks and decision-support applications*

Recent research has increasingly recognized that effective decision support requires more than accurate isolated models. In many application domains, decisions emerge from the interaction between automated units, expert knowledge, and contextual considerations that may change over time. For clarity, it is useful to distinguish two related but different bodies of work: general cooperative-decision frameworks, which address how multiple decision entities interact, and application-oriented decision-support sys-

tems, which implement concrete methods for a domain or family of decision problems.

At the framework level, cooperative game theory offers one formal perspective on how decision-making entities with heterogeneous roles or attributes may combine their contributions, including settings in which each agent is characterized by multiple attributes rather than a single payoff value (Borkotokey, Hazarika, & Mesiar, 2019). CADEMAs does not adopt this game-theoretic formulation, but shares its concern with making cooperation among heterogeneous entities explicit. At the operational level, Tariq, Chhetri, Nepal, and Paris (2024) propose an AI-to-collaborator (A2C) framework in which humans and AI systems share decision authority through explicit interaction mechanisms. Related work on hybrid intelligence emphasizes that authority distribution and human interpretive capacity should be modeled separately, because the level of AI integration and the cognitive flexibility of human decision-makers jointly shape decision quality and accountability (Deliu & Moldovan, 2026). A neighboring line of work examines cooperation among multiple AI agents, including LLM-based multi-agent reinforcement learning (Sun, Huang, & Pompili, 2024). These contributions provide relevant points of comparison because they address cooperation among decision entities. However, their main focus is not the reusable implementation of a CADEMAs-like workflow in which heterogeneous ADM outputs are explicitly separated from context modulation and final prioritization.

A related theoretical thread connects cooperative game theory to case-level explanation. The Shapley value distributes a joint outcome among players according to their averaged marginal contributions (Borkotokey et al., 2019) and has been adopted in explainable artificial intelligence as a principled attribution mechanism. SHapley Additive exPlanations (SHAP) (Lundberg & Lee, 2017) treat model features as players and decompose each prediction into additive feature contributions with axiomatic guarantees; recent surveys document the breadth of this cooperative-to-XAI bridge (Li, Sun, Huang, & Chen, 2024). This literature motivates the inclusion of an explicit explanation module in CADEMAs-ML. The current implementation does not compute Shapley values over features: because the uploaded H2O MOJO models are treated as black boxes and may include stacked ensembles, the tool uses a model-agnostic perturbation procedure instead. The purpose is nonetheless the same as in Shapley-inspired XAI: to make the local drivers of a recommendation inspectable alongside the cooperative and contextual scores.

At the application and decision-support level, substantial progress has been made in fuzzy decision support, multi-criteria decision making, and hybrid fuzzy-ML workflows. Applications such as MAKEDECISION (Więckowski & Sałabun, 2024) and the fuzzy DEMATEL-based tool of Chekry et al. (2024) emphasize transparency and traceability by exposing intermediate calculations to users. A related stream has explored hybrid pipelines that combine expert judgment, causal weighting, and predictive components in domain-specific settings (Abdulgader, Eid, & Daneshvar Rouyendegh, 2018;

Özkurt, Canay, Tunç, Aydin, & Velioğlu, 2025; Sahraei et al., 2022). Recent work has also extended the fuzzy toolbox through aggregation operators that account for interactions among decision attributes, capturing dependency structures that simple weighted averages cannot represent (Kacprzyk, Sirbiladze, & Tsulaia, 2022). The importance of explainability in AI-based decision support has been broadly recognized: any automated system whose outputs must be justified or audited requires transparent mechanisms to convey its reasoning to stakeholders (Barredo Arrieta et al., 2020), a requirement increasingly addressed through XAI techniques, including Shapley-value attribution (Li et al., 2024; Lundberg & Lee, 2017), in hybrid ranking and prediction pipelines (Özkurt et al., 2025). These contributions demonstrate the value of combining data-driven evidence with contextual or expert-defined criteria, but most are tailored to particular methods or domains rather than organized around the CADEMAs decomposition of ADM units, integration, context, and final prioritization.

Open-source fuzzy and MCDM libraries, including `pyFDM` (Więckowski, Kizielewicz, & Sałabun, 2022), `PyIFDM` (Więckowski, Kizielewicz, & Sałabun, 2023), the `IFMCDM` package of Roszkowska, Jefmański, Dudek, and Kusterka-Jefmańska (2024), the `ELIGERE` platform (Grazioso, Caporaso, & di Gironimo, 2023), and the intuitionistic fuzzy dimensional-analysis approach of Pérez-Domínguez, Callejas-Cuervo, and García-Luna (2024), are also relevant. They improve reproducibility and provide reusable computational building blocks.

Table 1 summarizes representative application-oriented contributions in these areas. Three observations emerge from this comparison. First, published systems already provide strong ingredients for transparent decision support, including human–AI collaboration, auditable fuzzy MCDM, hybrid ranking pipelines, and reusable software libraries (Özkurt et al., 2025; Tariq et al., 2024; Więckowski et al., 2022; Więckowski & Sałabun, 2024). Second, these ingredients are usually developed in isolation. Third, recent CADEMAs applications already combine cooperative multi-ADM integration with an explicit contextual layer (Godz et al., 2026; Novoa-Hernández et al., 2026; Vazquez-Sánchez et al., 2026), yet they remain domain-specific configurations rather than a reusable software workflow with case-level explanation. What remains missing is an operational implementation that keeps these capabilities configurable and auditable together.

CADEMAS-ML is designed to address this gap for the predictive-model setting. Building on the framework and its attrition instantiation discussed above, it provides a configurable software implementation in which ML-based ADM outputs, contextual reasoning, and case-level explanation coexist as explicit and auditable components of the decision-making process. The source code is publicly available under an MIT license (Section 5); the main contribution is the operationalization of the CADEMAs decomposition in a reusable decision-support tool.

Table 1. Comparison of representative application-oriented decision-support works related to CADEMAs-ML.

Work	Decision-support focus	Multiple ADM units	Explicit context layer	Case-level XAI	Relation to CADEMAs-ML
Tariq et al. (2024)	Human–AI collaborative decision framework with automated, augmented, and collaborative modes	Partly	Partly	No	Strong framework-level precedent, but not a configurable fuzzy/ML prioritization workflow
Więckowski and Sałabun (2024)	Auditable fuzzy MCDM through a web-based tool	No	Partly	No	Strong transparency, but no cooperative ADM integration layer
Chekry et al. (2024)	Open-source fuzzy DEMATEL decision-support tool	No	Partly	No	Traceable fuzzy causal analysis, but method-specific rather than CADEMAs-like
Özkurt et al. (2025)	Hybrid fuzzy MCDM, ML, and XAI for transparent ranking and prediction	Partly	Partly	Partly	Combines ranking, prediction, and XAI, but lacks a general cooperative ADM architecture
Sahraei et al. (2022)	Fuzzy DEMATEL/BWM/ANP pipeline for spatial risk prioritization	No	Partly	No	Explicit causal weighting under uncertainty, but application-specific
Godz et al. (2026)	Employee-attribution prevention using CADEMAs with departmental ML models	Yes	Yes	No	Direct CADEMAs application that motivates the implemented workflow
Novoa-Hernández et al. (2026)	Aggregation semantics for CADEMAs prioritization under alternative operators	Yes	Yes	No	Analyzes how linear, geometric, and minimum rules preserve or compensate contextual requirements
Vazquez-Sánchez et al. (2026)	Cooperative ADM with fuzzy contextual modulation for SME insolvency prediction	Yes	Yes	Yes	Extends CADEMAs-style cooperation to survival models with contextual modulation and XAI support
CADEMAs-ML (this work)	Configurable prioritization with ML-based ADM units, fuzzy context scenarios, and local case explanation	Yes	Yes	Yes	Operationalizes CADEMAs for reusable, explainable ML-driven decision support

3. Instantiation of CADEMAs for the Prioritization Problem

3.1. Problem definition and overall mapping

~~As previously noted in Section 1, prioritization problems in organizational settings arise when a set of cases must be ordered according to their relative need for attention,~~

~~intervention, or resource allocation. Examples include identifying employees who may require retention measures, allocating limited resources among clients or cases, and scheduling inspections according to risk and organizational priorities.~~ Section 1 motivates prioritization when attention, intervention, or resources are limited. This section now formalizes that problem. Let $\mathcal{X} = \{x_1, \dots, x_N\}$ denote the set of cases considered in a decision process, where each case x_i is characterized by a set of observable attributes. ~~The objective is not only to assign a score to each case, but to establish a priority ordering over $\mathcal{X}$.~~ The objective is to score the cases and establish a priority ordering over $\mathcal{X}$.

Let

$$P : \mathcal{X} \rightarrow [0, 1] \quad (4)$$

denote a priority function that assigns a normalized priority score to each case. This score induces a preference relation over the cases such that

$$x_i \succeq_P x_j \iff P(x_i) \geq P(x_j). \quad (5)$$

The corresponding ranking can be represented by a function

$$\pi_P : \mathcal{X} \rightarrow \{1, \dots, N\}, \quad (6)$$

where lower values indicate higher priority. In the absence of ties, the rank of a case is therefore

$$\pi_P(x_i) = 1 + |\{x_j \in \mathcal{X} : P(x_j) > P(x_i)\}|. \quad (7)$$

Thus, a prioritization configuration defines both a priority score for each case and an ordering of the cases induced by those scores.

~~In practical settings, however, the priority function may need to reflect more than a single source of evidence. Different predictive or decision-making components may capture complementary aspects of a case, while the relevance of their outputs may depend on the organizational, operational, or regulatory context in which the prioritization is performed. A case with a high predicted risk, for example, may not have the same priority under different organizational objectives or contextual conditions. The prioritization problem can therefore be viewed as the combination of multiple partial assessments with contextual considerations before the final ordering is established. This structure makes prioritization a natural application of the CADEMAs framework introduced in Section 2.1. CADEMAs provides an architecture in which heterogeneous Automated Decision-Making (ADM) units can contribute~~

partial decision signals, these signals can be integrated into a common decision representation, and the resulting proposal can subsequently be considered in relation to the relevant context. A priority function may draw on several assessments and produce different rankings under different organizational conditions. The integration and modulation stages defined in Section 2.1 provide the required structure. CADEMAs-ML specializes this architecture for the prioritization problem by implementing the ADM units as machine-learning models and providing explicit mechanisms for defining the data, model, and contextual components involved in the decision process. The resulting specialization can be summarized through the mapping

$$\mathcal{S} = \langle \mathcal{IM}, \mathcal{A}, \mathcal{OM}, \mathcal{C}, \mathcal{T} \rangle$$

as follows:

- **Data representation ($\mathcal{T}$):** The admissible data representations are specialized to the tabular case data and the decision signals required by the predictive and contextual components. This provides a common data basis while allowing individual ADM units and contextual models to use only the variables relevant to them. Tabular case data provide the variables and decision signals required by the predictive and contextual components. Each component may use its own subset.
- **Input Manager ($\mathcal{IM}$):** The Input Manager coordinates the information required by the different components of the system. In particular, it defines how case data are provided to the ADM units, which features are required by each predictive model, and which variables are available for contextual modeling. These requirements are specified independently of the execution of the models, allowing different predictive models, their reported performance measures, and contextual definitions to be incorporated without modifying the underlying decision process. The Input Manager coordinates case data, model requirements, performance metadata, and contextual definitions. These inputs can change without modifying the decision process.
- **ADM units ($\mathcal{A}$):** The ADM layer is realized through a collection of pre-trained machine-learning models that represent different predictive perspectives on the cases. Each model may use a different subset of the available attributes and contributes its own assessment of the target event. Their independent predictions constitute the partial decision signals that will subsequently be integrated. Pre-trained machine-learning models provide predictive perspectives on each case. A model may use its own attribute subset and contributes one partial signal.
- **Decision Integration Layer ($\mathcal{OM}$):** The integration layer combines the predictive assessments provided by the ADM units into a common representation

of predictive risk. The contribution of each model can be determined from its reported predictive performance, allowing models with different predictive quality to participate in the cooperative decision without requiring them to be retrained or structurally modified. The integration layer combines the model assessments into a predictive score. Reported performance determines each model's contribution.

- **Contextual Modulator ($\mathcal{C}$):** The contextual component represents the conditions under which the predictive evidence is interpreted. In CADEMAs-ML, contextual definitions specify which attributes of the input cases are relevant to a particular decision scenario and how their values contribute to contextual suitability. The resulting contextual assessment is then incorporated into the prioritization together with the integrated predictive evidence. Context definitions select relevant case attributes and evaluate their suitability for a scenario. The modulator combines this assessment with the predictive score.

The following subsections specify the realization of the predictive integration and contextual modulation, including the mechanisms used to configure and combine their respective contributions. The final subsection introduces the explainability and robustness extensions added to this specialization. The following subsections formalize these components and then introduce the explainability and robustness extensions.

3.2. *Input Manager, ADM Units and Data Representation*

In CADEMAs-ML, the Input Manager $\mathcal{IM}$ materializes the information required to configure and execute the different components of the prioritization process. The Input Manager $\mathcal{IM}$ groups the information needed to configure and execute a prioritization. It is represented as the tuple

$$\mathcal{IM} = \langle \mathcal{M}, \mathcal{M}_{def}, \mathcal{C}_{def}, \mathcal{X} \rangle, \quad (8)$$

where:

- $\mathcal{M} = M_1, \dots, M_K$ is the collection of pre-trained machine-learning models used as ADM units. Each model provides a partial predictive assessment of the cases and may rely on a different subset of the available features.
- $\mathcal{M}_{def} = m_1, \dots, m_K$ contains the corresponding model definitions. Each definition specifies the metadata required to use a model within the CADEMAs instance, including its identifying information, predictive performance measures, and the features required as input. Each definition records model identity, predictive performance, and input features.

- $\mathcal{C}_{def}$ is the collection of contextual definitions used to configure the contextual component of the instance. ~~A contextual definition specifies the case attributes considered relevant to a particular scenario and the corresponding mechanisms used to evaluate their contextual suitability.~~ Each definition identifies scenario-relevant attributes and their suitability rules.
- $\mathcal{X} = x_1, \dots, x_N$ is the set of cases to be prioritized. Each case is represented by a set of observable features sufficient to satisfy the input requirements of the configured models and contextual definitions, i.e.,

$$f(x_i) \supseteq \bigcup_{k=1}^K f_k \cup f_c, \quad (9)$$

where f_k denotes the features required by model M_k and f_c the features required by the configured contextual definition.

~~The data representation $\mathcal{T}$ is specialized to tabular case data and the structured representations required to configure models and contextual definitions. In particular, the case data provide the feature values consumed by the ADM units and contextual component, while model and context definitions provide the metadata and declarative specifications required for their execution.~~ The representation $\mathcal{T}$ comprises tabular case data and the structured model and context definitions required for execution.

~~It is worth noting that although these elements are defined as sets to support the evaluation of different configurations or scenarios, each CADEMAs-ML prioritization process operates on a specific set of models together with their metadata definitions, a single contextual definition, and a single set of case data. Each run selects one model collection with its metadata, one contextual definition, and one case dataset.~~

~~Thus, $\mathcal{IM}$ provides the data and configuration required by the subsequent CADEMAs components, with the main requirement that all features required by the configured models and contextual definition are present in the case data and mutually consistent. The corresponding file formats and examples are described in Section 4 and in the Supplementary Material. All features required by the selected models and context must be present and mutually consistent. Section 4, Appendix C, and Supplementary Material S1 describe the corresponding files.~~

3.3. *Decision Integration Layer*

~~The Decision Integration Layer $\mathcal{OM}$ materializes the cooperation among the machine-learning-based ADM units by synthesizing their individual predictive outputs into a common decision representation.~~ The Decision Integration Layer $\mathcal{OM}$ synthesizes the model outputs into the predictive score introduced in Section 3.1. For a given

prioritization scenario, each model $M_k \in \mathcal{M}$ produces a prediction

$$p_{ik} = M_k(x_i), \quad p_{ik} \in [0, 1], \quad (10)$$

for case x_i . ~~These individual predictions constitute the partial decision signals provided by the ADM units.~~ These predictions are the partial signals defined by the general architecture (Section 2.1).

To account for differences in predictive performance, each model is assigned a non-negative weight w_k , with

$$w_k \geq 0, \quad \sum_{k=1}^K w_k = 1. \quad (11)$$

The integrated decision signal for case x_i is then obtained as

$$D_i^{\text{raw}} = \Phi(p_{i1}, \dots, p_{iK}) = \sum_{k=1}^K w_k p_{ik}, \quad (12)$$

where $D_i^{\text{raw}} \in [0, 1]$ is the raw decision proposal resulting from the integration of the ADM outputs. ~~In the present prioritization application, this value can be interpreted as an integrated predictive score, whose subsequent interpretation and combination with contextual information are determined by the Contextual Modulator.~~ For prioritization, it is the integrated *predictive score* passed to the Contextual Modulator.

The weighting mechanism implemented by CADEMAs-ML is based on the reported predictive performance of the models and is described below. The integration operates on the outputs of the ADM units without modifying their internal models, allowing heterogeneous predictive models to contribute to a common decision representation. The weights are derived from reported predictive performance. Integration uses only model outputs and does not modify the models.

~~Although the resulting synthesis can technically be regarded as an ensemble-based prediction, CADEMAs-ML does not require the ADM units to have been trained jointly or with a common objective aimed at maximizing collective predictive performance. The models may therefore have been developed independently, using different training datasets, feature subsets, algorithms, or predictive objectives. Their cooperation takes place at the decision level, through the integration of their outputs within the CADEMAs architecture. This is decision-level cooperation, not a requirement for joint training. The models may differ in their training data, feature subsets,~~

algorithms, and objectives.

3.4. Contextual Modulator

~~The Contextual Modulator $\mathcal{C}$ materializes the contextual component of CADEMAs-ML through a declarative specification of fuzzy contextual conditions.~~ The Contextual Modulator $\mathcal{C}$ represents contextual conditions through a declarative fuzzy specification. Following the role of context in Novoa-Hernández et al. (2026) and the use of fuzzy propositions in the attrition application of Godz et al. (2026), each context definition specifies the case attributes to be considered, the corresponding fuzzy propositions, and the aggregation rules used to obtain a contextual assessment. ~~Unlike those application-specific implementations, CADEMAs-ML represents these definitions independently of the application code, allowing different contexts and decision policies to be configured through JSON specifications.~~ CADEMAs-ML externalizes these definitions as JSON, so contexts and policies can change independently of the application code.

Formally, let V_r denote a case attribute and $\tilde{F}_r$ a fuzzy set representing a linguistic predicate over the universe of V_r . For case x_i , an *atomic contextual rule* evaluates the proposition “ V_r is $\tilde{F}_r$ ” as

$$\mu_r(x_i) = \mu_{\tilde{F}_r}(x_{ir}) \in [0, 1], \quad (13)$$

where x_{ir} is the value of V_r for case x_i . ~~CADEMAs-ML supports triangular, trapezoidal, linear increasing, and linear decreasing membership functions, as well as categorical maps and categorical sets.~~ The supported numerical and categorical membership functions are specified in Appendix B.1.

Atomic rules may also include activation conditions. Let $m_r(x_i) \in \{0, 1\}$ indicate whether the **when** condition of rule r is satisfied for case x_i . Its effective membership is then

$$\mu_r^*(x_i) = m_r(x_i)\mu_r(x_i), \quad (14)$$

with $m_r(x_i) = 1$ for unconditional rules. This allows conditional policies to be expressed declaratively, without modifying the case data or introducing procedural logic.

Context definitions may further include *derived rules* that combine previously evaluated memberships. For a derived rule d , its membership is given by

$$\mu_d(x_i) = G_d \left(\mu_{r_1}^*(x_i), \dots, \mu_{r_q}^*(x_i) \right), \quad (15)$$

where G_d is an aggregation operator. ~~The implemented operators include minimum-based AND, maximum-based OR, algebraic PRODUCT, and arithmetic AVERAGE.~~ Appendix B.3 defines the implemented operators. The contextual alignment score for case x_i is finally obtained as

$$C_i = H(\mu_1(x_i), \dots, \mu_L(x_i)), \quad (16)$$

where H is the final aggregation policy specified by the context definition.

The resulting C_i represents the degree to which case x_i satisfies the contextual conditions of the selected scenario. ~~All atomic and derived memberships are retained by the tool, making the contextual assessment traceable to the underlying fuzzy rules rather than treating it as an opaque modification of the integrated predictive signal.~~ The tool retains every atomic and derived membership, so the assessment remains traceable to its rules.

3.4.1. Contextual Modulation

The contextual assessment C_i is subsequently used by the Contextual Modulator to transform the raw decision proposal D_i^{raw} into the final decision value used for prioritization. ~~In accordance with the general CADEMAs formulation, this modulation is not prescribed by the framework itself. CADEMAs-ML therefore implements three alternative aggregation operators for combining predictive and contextual evidence, namely the weighted arithmetic mean, the weighted geometric mean, and the minimum operator.~~ Within the freedom provided by the general formulation (Section 2.1), CADEMAs-ML offers weighted arithmetic, weighted geometric, and minimum operators.

Let $D_i^{\text{raw}} \in [0, 1]$ denote the integrated predictive signal and $C_i \in [0, 1]$ the contextual alignment score. The final decision value is obtained as

$$D_i^{\text{final}} = M(D_i^{\text{raw}}, C_i). \quad (17)$$

For the weighted arithmetic operator, used in previous CADEMAs applications (Godz et al., 2026; Novoa-Hernández et al., 2026), the modulation is

$$D_i^{\text{final}} = \lambda D_i^{\text{raw}} + (1 - \lambda) C_i, \quad \lambda \in [0, 1]. \quad (18)$$

CADEMAS-ML additionally implements the weighted geometric operator,

$$D_i^{\text{final}} = (D_i^{\text{raw}})^\lambda (C_i)^{1-\lambda}, \quad (19)$$

and the minimum operator,

$$D_i^{\text{final}} = \min \{ D_i^{\text{raw}}, C_i \}. \quad (20)$$

The inclusion of these alternatives follows the analysis in Novoa-Hernández et al. (2026), which examined the behavior of different aggregation operators for combining predictive and contextual signals. In particular, the geometric and minimum operators provide more restrictive forms of cooperation than the arithmetic operator, reducing the possibility that a low predictive signal or a low contextual alignment is compensated by a high value of the other component. The parameter λ controls the relative contribution of predictive and contextual information for the arithmetic and geometric operators, whereas the minimum operator has no weighting parameter. The geometric and minimum operators restrict compensation between a low predictive score and low contextual alignment. The parameter λ controls their relative contribution in the arithmetic and geometric cases; the minimum operator has no weight.

3.5. *Explainability Extension*

CADEMAS-ML extends the original CADEMAs framework with a case-level explanation mechanism designed to make the cooperative decision process interpretable and auditable. While CADEMAs distinguishes the outputs of ADM units, their integration, and contextual modulation, it does not prescribe how these elements should be exposed to decision-makers. The proposed extension addresses this aspect by retaining intermediate computational artifacts and providing two complementary explanation layers: a predictive explanation, which identifies the local influence of the input features on the integrated predictive signal, and a contextual explanation, which traces the fuzzy evidence underlying the contextual assessment. Case-level explanation exposes the intermediate artifacts produced by the integration and contextual stages. A predictive layer estimates the local influence of input features on D_i^{raw} . A contextual layer traces the fuzzy evidence underlying C_i .

For a selected case x_i , the explanation preserves the aggregation mechanism used to obtain the final decision. The predictive and contextual component values, D_i^{raw} and C_i , are reported together with the selected aggregation operator and, when applicable, its parameter λ . ~~For the linear operator, the two components can be interpreted additively. For the weighted geometric operator, the explanation reflects their multiplicative combination and the corresponding balance parameter. For the minimum operator, the lower component is identified as the limiting factor. This representation preserves the semantics of each aggregation mechanism without imposing an additive decomposition on operators for which such a decomposition is not defined.~~ The explanation is additive for the linear operator and multiplicative for the geometric operator. For the minimum operator, it identifies the lower component as the limiting factor.

To explain the predictive component, CADEMAs-ML employs a model-agnostic perturbation procedure suitable for the heterogeneous and potentially ensemble-based ML models supported by the tool. For each model M_k and feature f , the local effect is estimated by comparing the original prediction with the prediction obtained after replacing the observed feature value with a cohort reference value b_f :

$$\Delta_{ikf} = M_k(x_i) - M_k\left(x_i^{(f \leftarrow b_f)}\right), \quad (21)$$

where b_f is the median of the reference cohort for numerical features and the mode for categorical features. The model-specific effects are then integrated using the same weights employed by the Decision Integration Layer:

$$\Delta_{if} = \sum_{k=1}^K w_k \Delta_{ikf}. \quad (22)$$

Positive values indicate that the observed feature value increases the integrated predictive score relative to the reference value, whereas negative values indicate a decrease. ~~These quantities provide approximate local feature effects in probability points and are not interpreted as causal effects.~~ They are approximate local effects in probability points, not causal estimates. The resulting predictive explanation can therefore be represented as

$$\mathcal{E}_i^{\text{pred}} = \{(f, \Delta_{if})\}_{f \in \mathcal{F}}, \quad (23)$$

where $\mathcal{F}$ denotes the set of features considered in the perturbation analysis.

The contextual explanation traces the assessment C_i back to the fuzzy propositions

and derived rules defined in the selected context. By retaining the membership values of the evaluated rules, the tool exposes the intermediate evidence underlying the contextual aggregation:

$$\mathcal{E}_i^{\text{context}} = \{(r, \mu_r^*(x_i))\}_{r \in \mathcal{R}_a} \cup \{(d, \mu_d(x_i))\}_{d \in \mathcal{R}_d}, \quad (24)$$

where $\mathcal{R}_a$ and $\mathcal{R}_d$ denote the sets of atomic and derived rules evaluated for case x_i , respectively. The use of $\mu_r^*(x_i)$ for atomic rules accounts for possible activation conditions. ~~This representation allows the contextual assessment to be inspected in terms of the conditions satisfied by the case, including conditional rule activations and the aggregation operations leading to C_i .~~ The retained values connect C_i to the conditions satisfied by the case.

The two explanation layers are subsequently combined into a deterministic natural-language representation. Formally, the explanation for case x_i can be represented as

$$E_i = \Psi \left(D_i^{\text{final}}, D_i^{\text{raw}}, C_i, \mathcal{E}_i^{\text{pred}}, \mathcal{E}_i^{\text{context}}, \mathcal{A}_i \right), \quad (25)$$

where $\mathcal{A}_i$ denotes the selected aggregation mechanism and its parameters when applicable. ~~The function Ψ translates the quantitative evidence into a structured textual explanation that relates the final prioritization outcome to its predictive and contextual drivers. In this way, the natural-language layer does not replace the underlying explanation artifacts, but provides a more accessible interpretation of how the predictive evidence, contextual suitability, and selected cooperation mechanism jointly determine the final decision.~~ The function Ψ translates this evidence into a structured account of the predictive and contextual drivers. The quantitative artifacts remain available for inspection.

3.6. Robustness Extension

~~CADEMAS-ML extends the CADEMAS decision process with a robustness analysis aimed at assessing the stability of prioritization results under alternative contextual modulation configurations.~~ As anticipated in Section 1, robustness analysis evaluates the stability of priorities across modulation configurations. Let

$$\mathcal{G}_{\text{mod}} = \{g = (o, \theta_o)\}$$

denote the set of modulation configurations explored by the analyst, where o identifies the selected aggregation operator and θ_o denotes its associated parameterization, which

may be empty for parameter-free operators. Each configuration $g \in \mathcal{G}_{\text{mod}}$ induces a prioritization score vector

$$\mathbf{D}^{(g)} = \left(D_1^{(g)}, \dots, D_N^{(g)} \right)$$

and an associated ranking $\pi^{(g)}$ over the cases.

~~This representation allows robustness to be examined from two complementary perspectives. Within-operator robustness evaluates the stability of prioritization results under different parameterizations of the same aggregation operator. For operators with a balance parameter, such as the linear and weighted geometric operators, the analyst can explore a set of λ values and compare the resulting score vectors and rankings. The minimum operator provides a parameter-free configuration that can be included as a reference in the broader analysis.~~ Within-operator robustness compares parameterizations of one operator. For linear and weighted geometric aggregation, the analyst varies λ and compares the resulting scores and rankings. The parameter-free minimum operator can serve as a reference.

~~Across-operator robustness evaluates the stability of prioritization results when the aggregation mechanism itself is changed. Given the same predictive and contextual signals, different operators in the set~~ Across-operator robustness changes the aggregation mechanism while holding the predictive and contextual signals fixed. The available operators form the set

$$\mathcal{O} = \{\text{linear, geometric, minimum}\}$$

~~can generate different score vectors and induced rankings. Comparing these configurations makes it possible to identify cases whose prioritization remains stable across alternative modulation mechanisms and cases whose scores or ranking positions are sensitive to the selected operator.~~ and may induce different scores and rankings. Their comparison identifies stable and operator-sensitive cases.

More generally, the tool considers the collection

$$\mathcal{R}_{\mathcal{G}} = \left\{ \pi^{(g)} : g \in \mathcal{G}_{\text{mod}} \right\}$$

of rankings induced by the explored configurations. ~~This collection can be inspected either individually or simultaneously, allowing the analyst to examine changes in the relative ordering of cases, shifts in their prioritization scores, and changes in the cases occupying the highest-priority positions. The same framework can therefore be used to perform within-operator and across-operator comparisons independently or jointly. The analyst can inspect these rankings individually or jointly and combine within-operator and across-operator comparisons.~~

~~The robustness analysis does not assume that one aggregation mechanism provides a universally preferred prioritization. Instead, it exposes the extent to which prioritization conclusions depend on the contextual modulation configuration. In particular, it allows the analyst to distinguish cases whose prioritization is stable across configurations from cases whose position is sensitive to changes in the balance between predictive evidence and contextual suitability or to the aggregation mechanism itself.~~ The analysis shows how strongly the conclusions depend on the balance between predictive and contextual evidence and on the selected operator.

4. Architecture and main functionalities

4.1. *Software architecture*

4.2. *Main functionalities*

4.2.1. *Demonstration setup: Employee-attribution case study*

4.2.2. *Integrated prioritization overview*

4.2.3. *Cooperative predictive risk audit*

4.2.4. *Contextual alignment evaluation*

4.2.5. *Case-level explanation*

4.2.6. *Robustness and sensitivity analysis*

5. Discussion

This paper presented CADEMAs-ML as an executable specialization of the CADEMAs framework for cooperative, context-aware, ML-based prioritization. Its contribution is not a new attrition predictor, but an operational separation of predictive evidence, contextual evaluation, and final prioritization. This separation builds on the general architecture of Novoa-Hernández et al. (2026) and the employee-attribution instantiation of Godz et al. (2026), while adding reusable configuration, case-level explanation, and ranking-sensitivity analysis. The employee-attribution case study shows the practical consequence of that design: rankings based on cooperative predictive risk need not coincide with rankings based on contextual alignment, and the explicit combination of both can be inspected rather than treated as an opaque post-processing step.

The tool also occupies a distinct position relative to related decision-support software. Existing platforms and libraries have made important advances in exposing intermediate fuzzy or multi-criteria calculations (Chekry et al., 2024; Więckowski &

Salabun, 2024), while hybrid approaches have combined prediction, ranking, expert knowledge, and explainability in application-specific pipelines (Özkurt et al., 2025). CADEMAs-ML complements these contributions by organizing the workflow around cooperating ADM units and by keeping their integrated output separate from the context in which a decision is made. The *Models*, *Context*, and *Overview* views preserve this distinction; the *Explain* and *Robustness* views then support case-level interpretation and policy-sensitivity analysis. These additions extend the static prioritization workflow considered in earlier CADEMAs work.

For practitioners, the main benefit of this architecture is the ability to revise decision policy without embedding that policy in the predictive models themselves. When organizational conditions change, analysts can retain the predictive layer, provide a different context specification, and examine how the resulting priorities change. The separate reporting of R_i , C_i , and $P_i(\lambda)$ makes the source of a recommendation visible, while the λ sweep identifies cases whose position is stable and cases whose priority depends strongly on the selected balance. This does not remove the need for professional judgment; rather, it provides a structured record of the predictive and policy assumptions presented to the decision-maker. Potential applications include alert triage, review queues, retention programs, and allocation of limited support or inspection resources, provided that models, contextual rules, and decision thresholds are validated for the domain concerned.

For researchers, CADEMAs-ML provides a reproducible environment in which alternative cooperation, context-representation, explanation, and robustness mechanisms can be compared within a common architecture. The current implementation uses performance-normalized weights to combine independently trained models and a convex linear rule, $P_i(\lambda) = \lambda R_i + (1 - \lambda)C_i$, to integrate predictive risk and context. This final rule is compensatory: sufficiently strong evidence in one component can offset weaker evidence in the other. Recent work on aggregation semantics for CADEMAs (Novoa-Hernández et al., 2026) motivates extending the software with operator families that express different compensation requirements, including conjunctive or veto-like behavior. Making the aggregation semantics configurable would allow the software to distinguish settings in which compensation is acceptable from those in which a minimum level of predictive or contextual support is required.

A related direction is to move cooperation upstream, from post-hoc output aggregation to model development. At present, the predictive ADM units are trained independently and cooperation begins when their probabilities are weighted. Future versions could investigate diversity-aware model selection, theory-informed learning constraints, or multi-objective procedures that balance predictive performance, calibration, and complementarity. The aim would not be to encourage disagreement for its own sake, but to obtain models that remain individually reliable while contributing non-redundant evidence from different variables, data sources, or perspectives on the

problem. Measures of conditional competence and redundancy could then inform both training and aggregation, providing a more principled trade-off between the performance of individual ADM units and the informational value of their cooperation.

Several limitations delimit these implications. The ADM units currently supported at run time are H2O MOJO classifiers; the broader CADEMAs notion of heterogeneous units, including optimization models, simulators, or rule-based systems, is not yet implemented. Local ML attributions are model-agnostic perturbations around a cohort baseline and should not be read as causal effects or as substitutes for model-specific explanation methods when model internals are available. The attrition case study uses a 20-case illustrative subset and is neither a predictive-performance study nor evidence for general human-resources policy. Moreover, context specifications require domain-expert elicitation, and the interface does not yet provide versioned approval workflows or immutable audit trails for production governance. Future development should therefore combine broader ADM support and richer aggregation semantics with empirical validation in additional domains, stronger context governance, and evaluation with the practitioners who would interpret and act on the resulting priorities.

Funding

This research was supported by the project “Study, Analysis and Evaluation of Cooperative Automated Decision-Making Systems (CADEMAS)”, PID2023-146575NB-I00, funded by MCIU/AEI/10.13039/501100011033 and by FEDER, EU.

Disclosure statement

The authors report there are no competing interests to declare.

Declaration of generative AI use

Generative AI tools were used to assist with language editing and manuscript formatting. The authors reviewed and take full responsibility for the content of this article.

Data availability statement

~~The illustrative employee records used in this article are a 20-case subset derived from the publicly available IBM HR Analytics Employee Attrition and Performance dataset~~
The case study uses a 20-record subset of the public IBM HR Analytics Employee Attrition and Performance dataset (Subhash, 2017). ~~The subset, model metadata, MOJO archives, and context specifications are distributed with the software repository cited in~~

~~Section 5.~~ Appendix A describes the subset. The data, metadata, models, and contexts are distributed with the software (Section 5).

Software availability

CADEMAS-ML is available at <https://github.com/pnovo/cademas-app> under an MIT license (Novoa-Hernández & Pelta, 2026). An archival snapshot will be deposited in Zenodo at the time of submission; the corresponding DOI will be inserted when it is available.

Supplementary material

Supplementary Material S1 accompanies this article and provides the complete specification of the four application inputs, field-level constraints, cross-file consistency requirements, and minimal configuration examples.

Author contributions

Pavel Novoa-Hernández: conceptualization, methodology, software, validation, writing – original draft, visualization. David A. Pelta: conceptualization, methodology, supervision, writing – review and editing.

References

- Abdulgader, F. S., Eid, R., & Daneshvar Rouyendegh, B. (2018, November). Development of decision support model for selecting a maintenance plan using a fuzzy MCDM approach: A theoretical framework. *Applied Computational Intelligence and Soft Computing*, 2018, 1–14.
- Barredo Arrieta, A., Díaz-Rodríguez, N., Del Ser, J., Bennetot, A., Tabik, S., Barbado, A., . . . Herrera, F. (2020, June). Explainable Artificial Intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI. *Information Fusion*, 58, 82–115.
- Borkotokey, S., Hazarika, P., & Mesiar, R. (2019, September). Cooperative games with multiple attributes. *International Journal of General Systems*, 48(8), 825–842.
- Bui, T., & Lee, J. (1999). An agent-based framework for building decision support systems. *Decision Support Systems*, 25(3), 225–237.
- Chekry, A., Bakkas, J., Hanine, M., Montero, E. C., Garat de Marin, M. S., & Ashraf, I. (2024). PyDEMATEL: A Python-based tool implementing DEMATEL and fuzzy DEMATEL methods for improved decision making. *SoftwareX*.

- Deliu, D., & Moldovan, A.-L. (2026). Human–AI collaboration and cognitive flexibility in industry 6.0: A framework for purposeful decision-making in innovation and project management. *International Journal of Managing Projects in Business*.
- Godz, M., Novoa-Hernández, P., & Pelta, D. A. (2026). A cooperative and context-aware approach for employee attrition prevention. In *Information processing and management of uncertainty in knowledge-based systems* (pp. 203–217). Springer Nature Switzerland.
- Grazioso, S., Caporaso, T., & di Gironimo, G. (2023). Decision support in engineering design: the ELIGERE open source software platform. *International Journal on Interactive Design and Manufacturing (IJIDeM)*.
- Herrera-Viedma, E., Herrera, F., & Chiclana, F. (2002). A consensus model for multiperson decision making with different preference structures. *IEEE Transactions on Systems, Man, and Cybernetics—Part A: Systems and Humans*, 32(3), 394–402.
- Kacprzyk, J., Sirbiladze, G., & Tsulaia, G. (2022). Associated fuzzy probabilities in MADM with interacting attributes: Application in multi-objective facility location selection problem. *International Journal of Information Technology & Decision Making*, 21(4), 1155–1188.
- Lamata, M. T., Pelta, D. A., & Verdegay, J. L. (2020, oct). The role of the context in decision and optimization problems. In *Fuzzy approaches for soft computing and approximate reasoning: Theories and applications* (pp. 75–84). Springer International Publishing.
- Li, M., Sun, H., Huang, Y., & Chen, H. (2024). Shapley value: From cooperative game to explainable artificial intelligence. *Autonomous Intelligent Systems*, 4(1), 2.
- Lundberg, S. M., Erion, G., Chen, H., DeGrave, A., Prutkin, J. M., Nair, B., ... Lee, S.-I. (2020). From local explanations to global understanding with explainable AI for trees. *Nature Machine Intelligence*, 2(1), 56–67.
- Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. In *Advances in neural information processing systems* (Vol. 30, pp. 4765–4774). Curran Associates. Retrieved from <https://proceedings.neurips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html>
- Navarro-Galera, A., Lara-Rubio, J., Novoa-Hernández, P., & Cruz Corona, C. A. (2025, March). Using decision trees to predict insolvency in spanish smes: Is early warning possible? *Computational Economics*, 65(1), 91–116. Retrieved from <http://dx.doi.org/10.1007/s10614-024-10586-5>
- Novoa-Hernández, P., Godz, M., & Pelta, D. A. (2026). Aggregation semantics for prioritization in cooperative automated decision-making. *Mathematics*, 14(18). Retrieved from <https://www.mdpi.com/2227-7390/14/18/3397>
- Novoa-Hernández, P., & Pelta, D. A. (2026). *CADEMAS-ML: A software tool for context-aware cooperative automated decision-making*. Computer software. (MIT License. <https://github.com/pnovoa/cademas-app>)
- Novoa-Hernández, P., Pelta, D. A., Godz, M., Verdegay, J. L., & Buendia-Carrillo, D. (2026). Cademas – a framework for cooperative automated decision-making systems. In *2026 ieee conference on artificial intelligence (cai)* (p. 756–761).
- Özkurt, C., Canay, O., Tunç, E. A., Aydin, E., & Velioğlu, B. S. (2025). Renewable energy source ranking and analysis using fuzzy MCDM, ML, and XAI techniques. *Baltic Journal*

- of *Modern Computing*, 13(3), 656–679.
- Pérez-Cañedo, B., Porras, C., Pelta, D. A., & Verdegay, J. L. (2023, may). Modeling contexts as fuzzy propositions in optimization problems. *IEEE Transactions on Fuzzy Systems*, 31(5), 1474–1483.
- Pérez-Domínguez, L., Callejas-Cuervo, M., & García-Luna, F. (2024). Intuitionistic fuzzy dimensional analysis in multi-criteria decision-making: A computational approach. *IEEE Access*, 12, 1–15.
- Roszkowska, E., Jefmański, B., Dudek, A., & Kusterka-Jefmańska, M. (2024). IFMCDM: An R package for intuitionistic fuzzy multi-criteria decision making methods. *SoftwareX*, 26, 101676.
- Sahraei, R., Kanani-Sadat, Y., Homayouni, S., Safari, A., Oubennaceur, K., & Chokmani, K. (2022). A novel hybrid GIS-based multi-criteria decision-making approach for flood susceptibility analysis in large ungauged watersheds. *Journal of Flood Risk Management*, 16(2), e12871.
- Salo, A. A. (1995). Interactive decision aiding for group decision support. *European Journal of Operational Research*, 84(1), 134–149.
- Subhash, P. (2017). *IBM HR analytics employee attrition and performance*. Kaggle dataset. (<https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset> (accessed 19 March 2026))
- Sun, C., Huang, S., & Pompili, D. (2024). LLM-based multi-agent reinforcement learning: Current and future directions. *AI*, 5(3), 1234–1252.
- Tariq, S., Chhetri, M. B., Nepal, S., & Paris, C. (2024). A2C: A modular multi-stage collaborative decision framework for human-AI teams. *Expert Systems with Applications*, 238, 122334.
- Vahidov, R., & Fazlollahi, B. (2004). Pluralistic multi-agent decision support system: A framework and an empirical test. *Information & Management*, 41(7), 883–898.
- Vazquez-Sánchez, A. A., Buendía-Carrillo, D., & Corona, C. C. (2026). A cooperative automated decision-making strategy with fuzzy contextual modulation for SME insolvency prediction. *Mathematics*, 14(19). Retrieved from <https://www.mdpi.com/2227-7390/14/19/3516>
- Więckowski, J., Kizielewicz, B., & Sałabun, W. (2022, December). pyFDM: A Python library for uncertainty decision analysis methods. *SoftwareX*, 20, 101271.
- Więckowski, J., Kizielewicz, B., & Sałabun, W. (2023, May). Handling decision-making in intuitionistic fuzzy environment: PyIFDM package. *SoftwareX*, 22, 101344.
- Więckowski, J., & Sałabun, W. (2024). MakeDecision: Online system for the graphical design of decision-making models in crisp and fuzzy environments. *SoftwareX*.

Notes on contributors

Pavel Novoa-Hernández is Associate Professor in the Department of Computer Engineering and Systems at Universidad de La Laguna, Spain. His research interests include automated decision-making, decision-making under uncertainty, evolutionary

computation, and machine learning applications.

David A. Pelta is Full Professor in the Department of Computer Science and Artificial Intelligence at Universidad de Granada, Spain. His research focuses on fuzzy optimization, evolutionary computation, and decision-making under uncertainty.

Appendix A. Attrition example dataset and model features

The 20-case attrition example distributed with CADEMAs-ML contains 36 columns: `Case_ID`, 34 predictor variables, and the `attrition` outcome label. ~~The predictors cover demographic, professional, compensation, performance, and organizational attributes from the IBM HR Analytics Employee Attrition and Performance Kaggle dataset used in~~ They describe demographic, professional, compensation, performance, and organizational characteristics from the dataset used by Godz et al. (2026). Five variables (`age`, `gender`, `department`, `job_role`, and `job_level`) are shared by all four MOJO models. Two additional columns (`employee_count` and `employee_number`) are retained in the CSV file but are not used by any model in the example bundle. ~~Each MOJO receives the full case matrix and selects its required variables according to the schema stored in the model-definition JSON file.~~ Each MOJO selects its variables from the full case matrix according to the model-definition schema.

Table A1 summarizes the feature assignment. The Human Resources model reads `over18` from the CSV file; the corresponding field is named `over_18` in the model metadata.

Appendix B. Context specification for the *Digital / Technological Transformation* scenario

The example repository includes two context configurations, but the walkthrough focuses on *Digital / Technological Transformation*. ~~This appendix reports the corresponding fuzzy specification in closed form so that every membership degree and the final context-alignment score C_i can be audited without inspecting application code. The rules instantiate the generic formalism of Equations (13)–(16); equivalent parameters are stored in the repository JSON file consumed by CADEMAs-ML~~ This appendix instantiates Equations (13)–(16) in closed form for that scenario. The equivalent JSON specification is distributed with CADEMAs-ML¹. ~~They are illustrative policy encodings, not hard-coded assumptions of the tool.~~ The policy is illustrative and remains separate from the application code.

¹https://github.com/pnovo/cademas-app/blob/main/example_attrition/context/context_digital_transformation.json.

Table A1. Predictor variables in the attrition example and the MOJO models that use each one. M_1 – M_4 denote Culture and Wellbeing, Finance, Operations and Development, and Human Resources, respectively.

Predictor	M_1 Culture & Wellbeing	M_2 Finance	M_3 Ops. & Dev.	M_4 Human Resources
age	✓	✓	✓	✓
business_travel	✓			
daily_rate		✓		
department	✓	✓	✓	✓
distance_from_home				✓
education				✓
education_field				✓
employee_count				
employee_number				
environment_satisfaction	✓			
gender	✓	✓	✓	✓
hourly_rate		✓		
job_involvement	✓			
job_level	✓	✓	✓	✓
job_role	✓	✓	✓	✓
job_satisfaction	✓			
marital_status				✓
monthly_income		✓		
monthly_rate		✓		
num_companies_worked			✓	
over18				✓
over_time	✓			
percent_salary_hike		✓		
performance_rating			✓	
relationship_satisfaction	✓			
standard_hours				✓
stock_option_level		✓		
total_working_years			✓	
training_times_last_year			✓	
work_life_balance	✓			
years_at_company			✓	
years_in_current_role			✓	
years_since_last_promotion			✓	
years_with_curr_manager			✓	
Total features	12	11	13	11

B.1. Generic membership functions

Each numerical atomic rule is evaluated through one of four parametric membership functions. For an input value x , the *linear increasing* function with breakpoints (a, b) , $a < b$, is

$$\mu_{a,b}^{\uparrow}(x) = \begin{cases} 0, & x \leq a, \\ \frac{x-a}{b-a}, & a < x < b, \\ 1, & x \geq b, \end{cases} \quad (\text{B1})$$

and the *linear decreasing* function with breakpoints (a, b) is its mirror image

$$\mu_{a,b}^{\downarrow}(x) = 1 - \mu_{a,b}^{\uparrow}(x) = \begin{cases} 1, & x \leq a, \\ \frac{b-x}{b-a}, & a < x < b, \\ 0, & x \geq b. \end{cases} \quad (\text{B2})$$

The *triangular* function with support (a, c) and peak b , $a < b < c$, is

$$\text{tri}_{a,b,c}(x) = \begin{cases} \frac{x-a}{b-a}, & a < x \leq b, \\ \frac{c-x}{c-b}, & b < x < c, \\ 0, & \text{otherwise,} \end{cases} \quad (\text{B3})$$

and the *trapezoidal* function with feet (a, d) and shoulders (b, c) , $a < b \leq c < d$, is

$$\Pi_{a,b,c,d}(x) = \begin{cases} \frac{x-a}{b-a}, & a < x < b, \\ 1, & b \leq x \leq c, \\ \frac{d-x}{d-c}, & c < x < d, \\ 0, & \text{otherwise.} \end{cases} \quad (\text{B4})$$

Categorical attributes are handled by a direct map $\kappa : \mathcal{S} \rightarrow [0, 1]$ that assigns a degree to each symbolic value $s \in \mathcal{S}$, with unspecified categories defaulting to 0.

B.2. Atomic rules

Seven atomic rules evaluate recent training intensity, education level, company tenure, tenure in the current role, age in the investment window, accumulated experience, and role adaptability to technological change. For case x_i , the corresponding memberships

are

$$\mu_{\text{train}}(x_i) = \mu_{1,6}^{\uparrow}(\text{training_times_last_year}_i), \quad (\text{B5})$$

$$\mu_{\text{edu}}(x_i) = \mu_{2,5}^{\uparrow}(\text{education}_i), \quad (\text{B6})$$

$$\mu_{\text{yrsCo}}(x_i) = \text{tri}_{3,10,20}(\text{years_at_company}_i), \quad (\text{B7})$$

$$\mu_{\text{yrsRole}}(x_i) = \text{tri}_{1,6,11}(\text{years_in_current_role}_i), \quad (\text{B8})$$

$$\mu_{\text{age}}(x_i) = \Pi_{22,28,50,57}(\text{age}_i), \quad (\text{B9})$$

$$\mu_{\text{exp}}(x_i) = \mu_{2,10}^{\uparrow}(\text{total_working_years}_i), \quad (\text{B10})$$

$$\mu_{\text{role}}(x_i) = \kappa_{\text{role}}(\text{job_role}_i), \quad (\text{B11})$$

where the role-adaptability map assigns expert scores to each job role:

$$\kappa_{\text{role}}(s) = \begin{cases} 0.9, & s \in \{\text{Research Scientist, Research Director}\}, \\ 0.8, & s = \text{Manager}, \\ 0.7, & s \in \{\text{Manufacturing Director, Healthcare Representative}\}, \\ 0.6, & s = \text{Sales Executive}, \\ 0.5, & s = \text{Human Resources}, \\ 0.4, & s \in \{\text{Laboratory Technician, Sales Representative}\}. \end{cases} \quad (\text{B12})$$

~~Each expression can be read directly: for instance, Equation (B9) states that ages between 28 and 50 fully satisfy the investment window ($\mu = 1$), the degree decays linearly toward the feet at 22 and 57, and ages outside [22, 57] are excluded. For example, Equation (B9) assigns full membership to ages from 28 to 50. Membership decreases linearly to zero at 22 and 57.~~

B.3. Derived rules and final aggregation

The aggregation operators of Equation (15), applied to a vector $\mathbf{u} = (u_1, \dots, u_q)$, are

$$\text{AND}(\mathbf{u}) = \min_j u_j, \quad \text{OR}(\mathbf{u}) = \max_j u_j, \quad \text{PROD}(\mathbf{u}) = \prod_{j=1}^q u_j, \quad \text{AVG}(\mathbf{u}) = \frac{1}{q} \sum_{j=1}^q u_j. \quad (\text{B13})$$

With these operators, three derived criteria combine the atomic memberships: *upskilling capacity* (training and education), *career scalability* (company and role tenure),

and *investment horizon* (age window and experience). Formally,

$$\mu_{\text{upskill}}(x_i) = \text{AVG}(\mu_{\text{train}}(x_i), \mu_{\text{edu}}(x_i)) = \frac{1}{2}(\mu_{\text{train}}(x_i) + \mu_{\text{edu}}(x_i)), \quad (\text{B14})$$

$$\mu_{\text{career}}(x_i) = \text{PROD}(\mu_{\text{yrsCo}}(x_i), \mu_{\text{yrsRole}}(x_i)) = \mu_{\text{yrsCo}}(x_i) \mu_{\text{yrsRole}}(x_i), \quad (\text{B15})$$

$$\mu_{\text{invest}}(x_i) = \text{PROD}(\mu_{\text{age}}(x_i), \mu_{\text{exp}}(x_i)) = \mu_{\text{age}}(x_i) \mu_{\text{exp}}(x_i). \quad (\text{B16})$$

~~The choice of operator carries a clear semantics: AVG lets strong training compensate for weaker formal education in *upskilling capacity*, whereas the PROD operator in *career scalability* and *investment horizon* is conjunctive, so a near-zero membership in either input collapses the derived degree toward zero. The context JSON also identifies four high-level criteria—upskilling, role adaptability, career scalability, and investment horizon—that can be evaluated as a declarative logic tree when no aggregation override is supplied.~~ The operators encode different relationships. In *upskilling capacity*, AVG allows stronger training to compensate for weaker education. The PROD operator is conjunctive, so a near-zero input reduces *career scalability* or *investment horizon* toward zero. When no aggregation override is supplied, the JSON logic tree combines four criteria: upskilling, role adaptability, career scalability, and investment horizon.

In the reported demonstration, the *average* operator is selected explicitly in the user interface. ~~This setting takes precedence over the declarative logic tree and averages all numeric atomic and derived memberships retained in the audit matrix.~~ It overrides the logic tree and averages the numeric atomic and derived memberships in the audit matrix. Let

$$\mathcal{S} = \{\text{train, edu, yrsCo, yrsRole, age, exp, role, upskill, career, invest}\}.$$

The context-alignment score used in the figures and numerical results is therefore

$$C_i = \frac{1}{|\mathcal{S}|} \sum_{r \in \mathcal{S}} \mu_r(x_i) = \frac{1}{10} \sum_{r \in \mathcal{S}} \mu_r(x_i). \quad (\text{B17})$$

~~Equation (B17) is the explicit instance of the generic policy in Equation (16) used for this run and defines the contextual evidence surfaced in the *Context* module (Figure reference pending) and traced at case level in the *Explain* module.~~ Equation (B17) instantiates the generic policy in Equation (16). Its evidence is available in the *Context* module and in each case-level explanation. Selecting *minimum (strict)* or *product* in the interface applies the corresponding operator to the same audit-matrix columns.

Appendix C. Input files and consistency requirements

~~CADEMAS-ML requires four coordinated inputs. Table C1 summarizes their roles~~

and principal constraints; Supplementary Material S1 gives the complete field-level specification and minimal examples. The specification distinguishes fields required for scoring from metadata used for display and explanation. Table C1 details the four inputs grouped by $\mathcal{IM}$ in Section 3.2. Supplementary Material S1 provides field-level constraints and examples, including the distinction between scoring fields and display or explanation metadata.

Table C1. Summary of the CADEMAs-ML input contract.

Input	Required content	Cross-file constraints
Case dataset (CSV)	One row per case and columns required by the uploaded models and context rules. A <code>Case_ID</code> or <code>ID</code> column is optional; otherwise, identifiers are generated.	Column names and data types must be compatible with the MOJO schemas. Every feature referenced by a context rule must be present. Identifier columns are excluded from inference.
Model metadata (JSON)	One top-level object per model, containing a <code>performance</code> object. The bundled schema also records <code>department</code> and <code>features</code> ; the latter is needed for the perturbation-based explanation workflow.	Each top-level key must exactly match an uploaded MOJO filename. The metric selected for weighting must be available for every participating model and should be a finite non-negative value.
Predictive models	H2O MOJO archives in <code>.zip</code> format that return compatible positive-class probabilities.	Each archive must have a matching metadata entry. All models must use the same positive-class interpretation, although their predictor subsets may differ.
Context specification (JSON)	A context name, atomic rules, optional <code>derived_rules</code> , and an optional final logic tree. Rules identify a feature, membership type, parameters, and optionally an activation condition.	Rule names must be unique, references in derived or final rules must resolve, and parameter shapes must match the membership type. The aggregation operator selected in the interface overrides the final JSON logic for that run.

~~The application checks filename matching and loads the selected metric at run time, but publication-quality deployments should validate the complete bundle before analysis. Such validation includes checking finite metric values, membership-parameter ordering, unresolved rule references, missing features, incompatible categorical values, and agreement on the positive class. Keeping these constraints explicit is important because the four files jointly define the executable decision model: none of them is fully interpretable in isolation.~~ At run time, the application checks filename matching and loads the selected metric. A deployment should also validate finite metrics, membership-parameter ordering, rule references, required features, categorical values, and positive-class consistency. These checks apply to the input bundle as a whole because its four files jointly define the executable decision model.